

Repair Approval Memo — RAM-2025-001

CASCADIA RENEWABLE OPERATIONS LLC

Internal Memorandum — Operations and Procurement

Field	Value
Document ID	RAM-2025-001
Date	2025-05-22
From	Priya Ramachandran, Plant Manager, NW-PowerPool
To	Daniel Voss, Director of Operations · cc: Lillian Park, CFO Office (cost flag review)
Subject	Approval — Apex Emergency Invoice EMR-2025-001 (WIN-10058 SoC drift event, \$412K)
EPA Reference	EPA-2024-11 (invoked 2025-05-02)
Distribution	Operations Leadership · CFO Office · Vendor Management · NW-PowerPool Records

1. Background

On 2025-05-02, WIN-10058 (GE wind turbine, NW-PowerPool-WIND-014) tripped on an inverter-side under-voltage event preceded by SoC drift on the co-located battery buffer. The forced outage ran 227 hours and produced **2,506.8 MWh of lost generation** (outage OUT-2025-0433). CPS could not mobilize within the 4-hour EPA-2024-11 window; the on-call duty manager invoked EPA and dispatched Apex Emergency Grid Solutions.

Apex completed onsite stabilization, partial capacitor replacement on the C-2 bank, and a BMM firmware re-flash. Restoration was achieved on 2025-05-12. The vendor's full report is **EMR-2025-001** and is on file.

This event is operationally similar to the 2025-04-11 SUB-10629 SoC drift event documented in **OIR-2025-001**. Both events trace back, in part, to the Q1 2025 budget freeze and the resulting deferral of preventive maintenance work orders at NW-PowerPool-WIND-014 and -024 (refer **CAR-2024-118**, **CAR-2025-002**).

2. Vendor Recommendation

Apex (EMR-2025-001, §4) recommends a **full inverter assembly and BMM replacement** within 120 days, with an estimated capital scope of \$1.05M (Apex proposal AP-Q-2025-118). Apex cites "irreversible degradation" of the DC-link and capacitor bank, referencing GE Service Advisory GE-SA-2022-14.

3. Cascadia Position

Operations agrees the emergency invoice (\$412K) reflects scope authorized under EPA-2024-11 and that the restoration met the immediate availability obligation. We do **not** agree, at this stage, with the full inverter/BMM replacement recommendation. Specifically:

- The Apex "irreversible degradation" characterization is consistent with Apex's pattern across NW-PowerPool deployments in FY2025 — every Apex EMR to date recommends a replacement-class follow-on. Reliability Engineering will independently assess the C-2 bank and BMM condition before any replacement decision.
- The proximate cause identified by Apex (capacitor degradation plus stale firmware) is consistent with deferred preventive maintenance. The asset-level remedy is to restore the preventive cadence, not necessarily to capitalize a \$1.05M replacement.
- This is the **4th Apex activation in FY2025** at NW-PowerPool (Dec 2024, April 2025, May 2025 — and a separate auxiliary). FY25 YTD Apex invoicing at NW-PowerPool now exceeds **\$1.4M**. The cumulative pattern warrants a Vendor Management review.

4. Financial Justification

- Invoice as presented: **\$412,000** (EMR-2025-001).
- Comparable scheduled scope (CPS retrospective estimate, 2025-04-22): **\$98,600**. The ratio (~4.2×) is consistent with the Apex-vs-CPS pattern observed across FY2025 emergency dispatches.
- NW-PowerPool FY25 maintenance budget impact: invoice will absorb ~3.1% of the \$13.4M annual line. After this invoice, NW-PowerPool emergency vendor spend YTD is approximately \$1.4M, on a path that will exceed the FY25 emergency reserve by Q3.
- Lost-generation revenue impact (separate from invoice): approximately \$150K at FY25 NW-PowerPool realized pricing.

5. Approval Request / Decision

Approved as presented, with the following conditions:

1. The \$412K Apex invoice EMR-2025-001 is approved for payment under EPA-2024-11.
2. The proposed \$1.05M inverter/BMM replacement (Apex AP-Q-2025-118) is **not** approved at this time. Reliability Engineering will perform an independent condition assessment and report under a CAR by Q3 close.
3. Vendor Management will open a formal Apex contract terms review, cross-referenced under **CAR-2025-004** (Apex performance and pricing review). Specific items: emergency response surcharge of 30%, risk/availability premium under MSA §4.3, and emergency parts mark-up of 35%.
4. Reliability Engineering and Procurement will jointly evaluate whether CPS Schedule B response can be hardened (e.g., positioned crew at NW-PowerPool) to reduce EPA-2024-11 invocations.

Approver	Role	Decision	Date
Priya Ramachandran	Plant Manager, NW-PowerPool	Approve invoice; recommend conditions	2025-05-22
Daniel Voss	Director of Operations	Approve	2025-05-23
Lillian Park	CFO Office (cost flag review)	Approve with note: flag cumulative Apex spend on monthly dashboard	2025-05-23

References: EMR-2025-001 · OIR-2025-001 · CAR-2024-118 · CAR-2025-002 · CAR-2025-004 (in preparation) · EPA-2024-11 · Apex MSA §4.3.